

BENCHMARK OVERVIEW

Ryzen AI MAX+ 392 vs 395 V5 Benchmark

OneNP controlled-setting rerun comparison

1) Benchmark scope

Models	Gemma 4 26B A4B / GPT-OSS 20B
Inputs	1K / 2K / 4K / 8K / 16K / 32K
Output / Trials	300 tokens / 3 runs
Runtime	llama.cpp build 10615 · Vulkan

2) Comparison modes

AI MAX+ 392	Reference result set
AI MAX+ 395 V5	Comparison result set
Δ convention	Positive = AI MAX+ 395 V5 faster
Model SHA256	Matched for both models

3) Metrics in this report

TTFT (ms)	Request start → first token
TPOT (ms/token)	Decode time per generated token
Aggregate throughput	Total output capacity (tok/s)
Normalized throughput	Per-user capacity (tok/s/user)

4) Reading guide

TTFT / TPOT	Lower is better
Throughput	Higher is better
NP1	Aggregate = Normalized
Interpretation	Observed benchmark deltas

METHOD NOTE Context policy and performance mode are aligned in V5: driver and Vulkan memory remain machine-specific, and threads use all logical processors.

Parameter	AI MAX+ 392	AI MAX+ 395 V5
Context	dynamic 4K-34.8K	dynamic 4K-34.8K
Threads	24	32
Power mode	Performance	High Performance + OEM Performance/Turbo
Driver	32.0.22032.3003	32.0.31035.1003
Vulkan memory	28,491 MiB	24,395 MiB

TTFT

TTFT

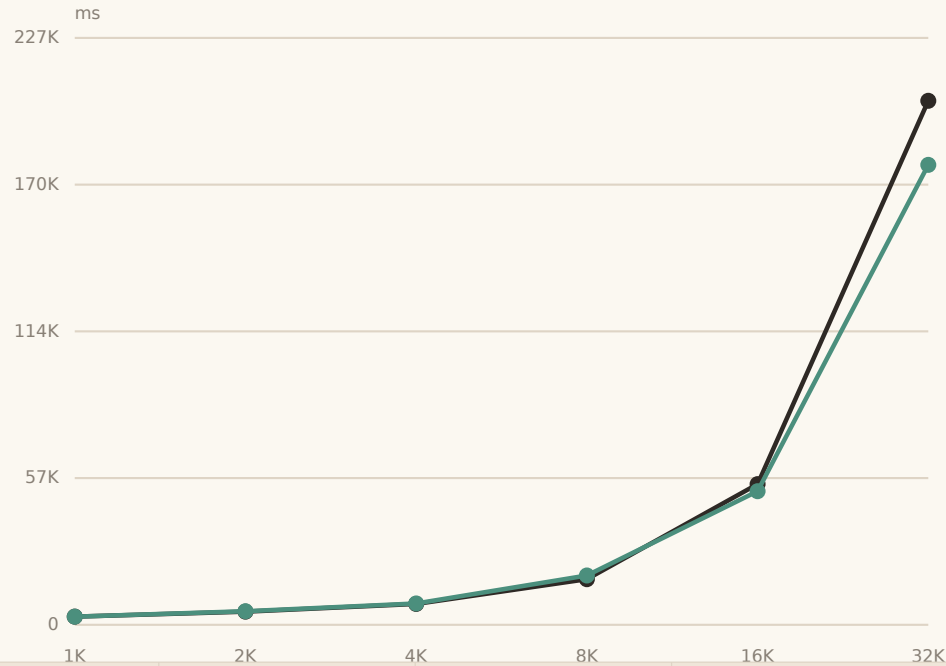
Time to first token (ms). Lower is better.

AI MAX+ 392 AI MAX+ 395 V5

FORMULA $\Delta (\%) = (\text{Reference} - \text{Comparison}) / \text{Reference} \times 100$
Positive means AI MAX+ 395 V5 measured faster

Gemma 4 26B A4B

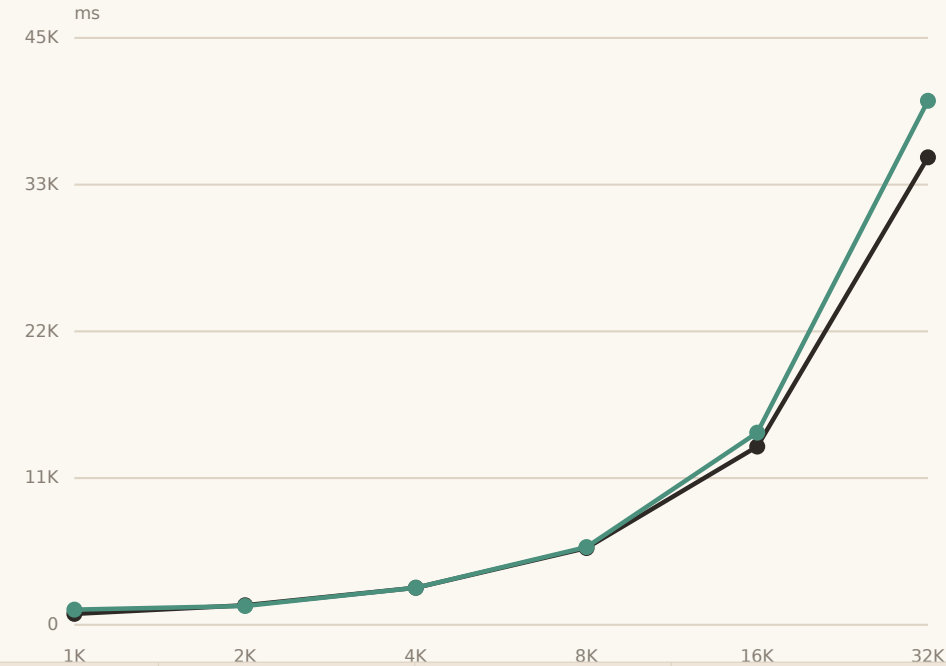
NP1 | Output 300



Input	AI MAX+ 392	AI MAX+ 395 V5	V5 Δ
1K	3,165	3,148	+0.5%
2K	5,087	5,256	-3.3%
4K	8,077	8,259	-2.3%
8K	17,658	19,095	-8.1%
16K	54,475	51,734	+5.0%
32K	202,915	178,107	+12.2%

GPT-OSS 20B

NP1 | Output 300



Input	AI MAX+ 392	AI MAX+ 395 V5	V5 Δ
1K	833	1,154	-38.5%
2K	1,485	1,430	+3.7%
4K	2,817	2,818	-0.0%
8K	5,841	5,912	-1.2%
16K	13,559	14,615	-7.8%
32K	35,559	39,859	-12.1%

KEY TAKEAWAY At 32K, AI MAX+ 395 V5 vs 392: Gemma +12.2%; GPT-OSS -12.1%.

TPOT

TPOT

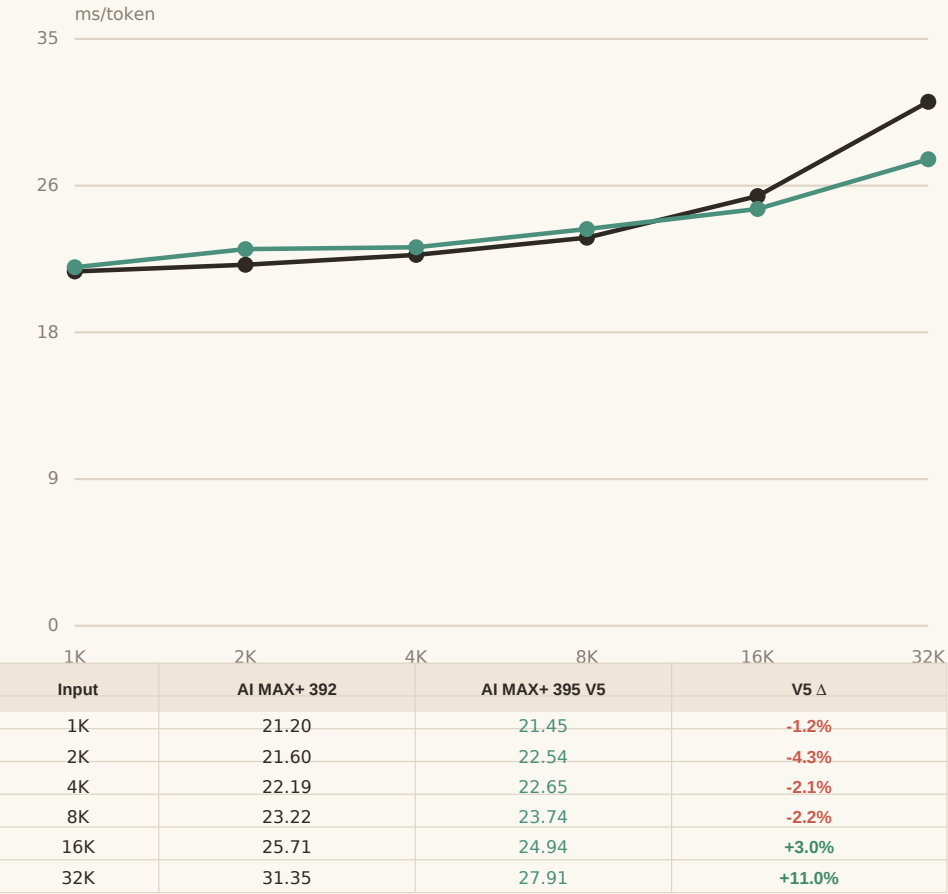
Median time per output token (ms/token). Lower is better.

AI MAX+ 392 AI MAX+ 395 V5

FORMULA $\Delta (\%) = (\text{Reference} - \text{Comparison}) / \text{Reference} \times 100$
Positive means AI MAX+ 395 V5 measured faster

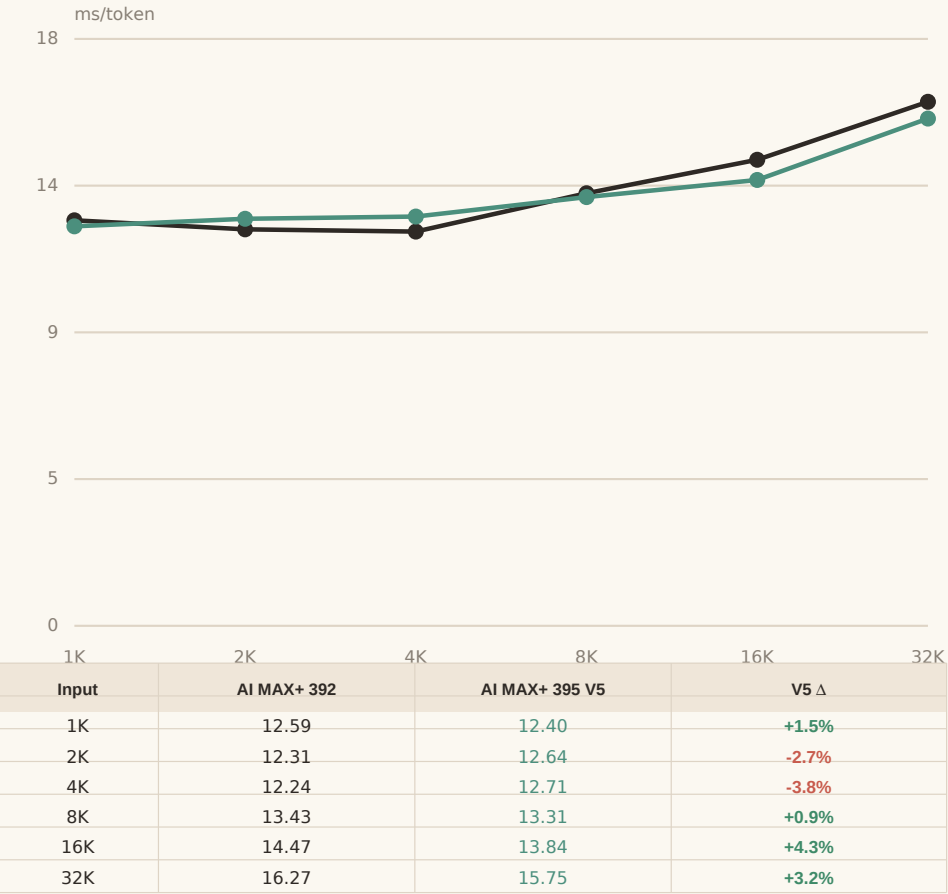
Gemma 4 26B A4B

NP1 | Output 300



GPT-OSS 20B

NP1 | Output 300



KEY TAKEAWAY At 32K, AI MAX+ 395 V5 vs 392: Gemma +11.0%; GPT-OSS +3.2%.

THROUGHPUT

Throughput (Aggregate)

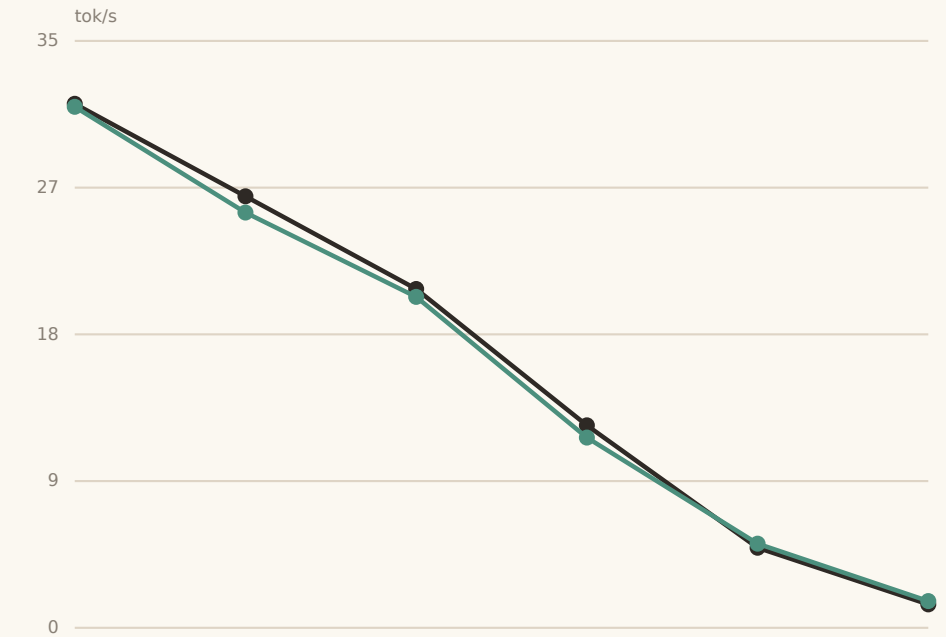
Total server output token throughput (tok/s). Higher is better.

AI MAX+ 392 AI MAX+ 395 V5

FORMULA $\Delta (\%) = (\text{Comparison} - \text{Reference}) / \text{Reference} \times 100$
Positive means AI MAX+ 395 V5 measured faster

Gemma 4 26B A4B

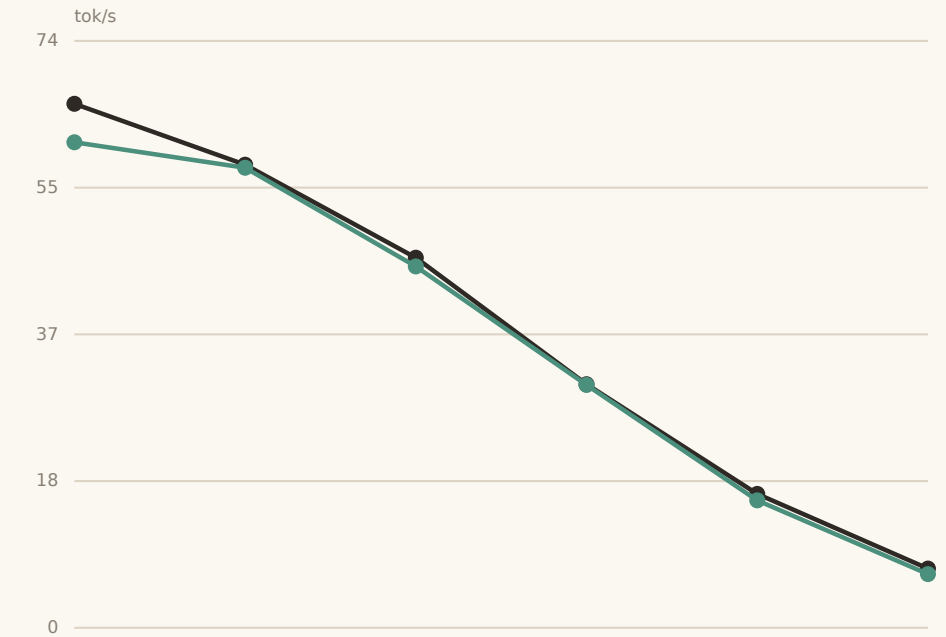
NP1 | Output 300



Input	AI MAX+ 392	AI MAX+ 395 V5	V5 Δ
1K	31.55	31.37	-0.6%
2K	25.98	25.00	-3.8%
4K	20.40	19.92	-2.3%
8K	12.19	11.45	-6.0%
16K	4.82	5.07	+5.1%
32K	1.41	1.61	+14.1%

GPT-OSS 20B

NP1 | Output 300



Input	AI MAX+ 392	AI MAX+ 395 V5	V5 Δ
1K	65.64	60.82	-7.3%
2K	58.01	57.63	-0.7%
4K	46.36	45.27	-2.4%
8K	30.51	30.42	-0.3%
16K	16.77	15.96	-4.8%
32K	7.42	6.73	-9.3%

KEY TAKEAWAY At 32K, AI MAX+ 395 V5 vs 392: Gemma +14.1%; GPT-OSS -9.3%.

THROUGHPUT

Throughput (Normalized)

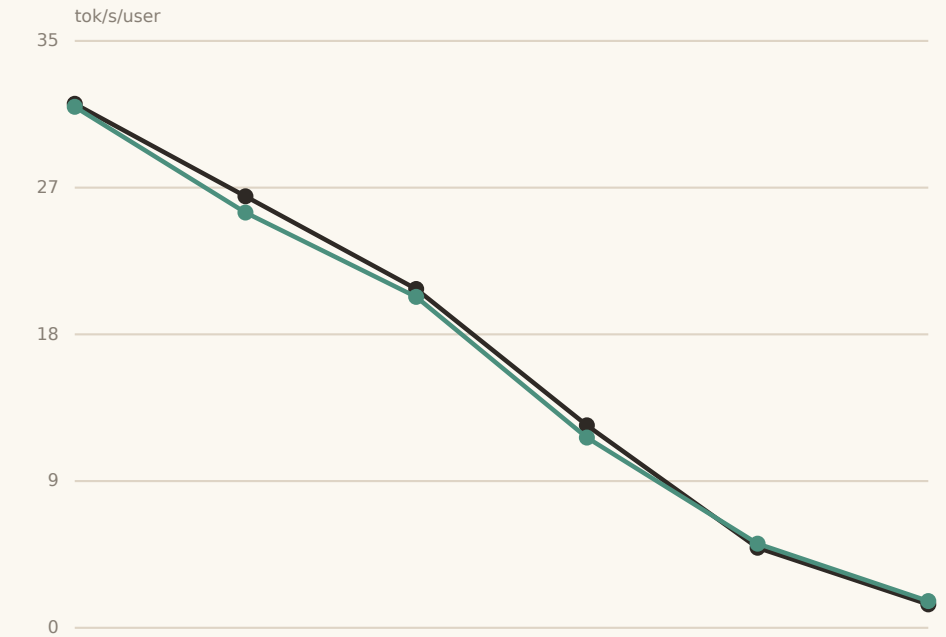
Aggregate throughput divided by NP (tok/s/user). Higher is better.

AI MAX+ 392 AI MAX+ 395 V5

FORMULA $\Delta (\%) = (\text{Comparison} - \text{Reference}) / \text{Reference} \times 100$
Positive means AI MAX+ 395 V5 measured faster

Gemma 4 26B A4B

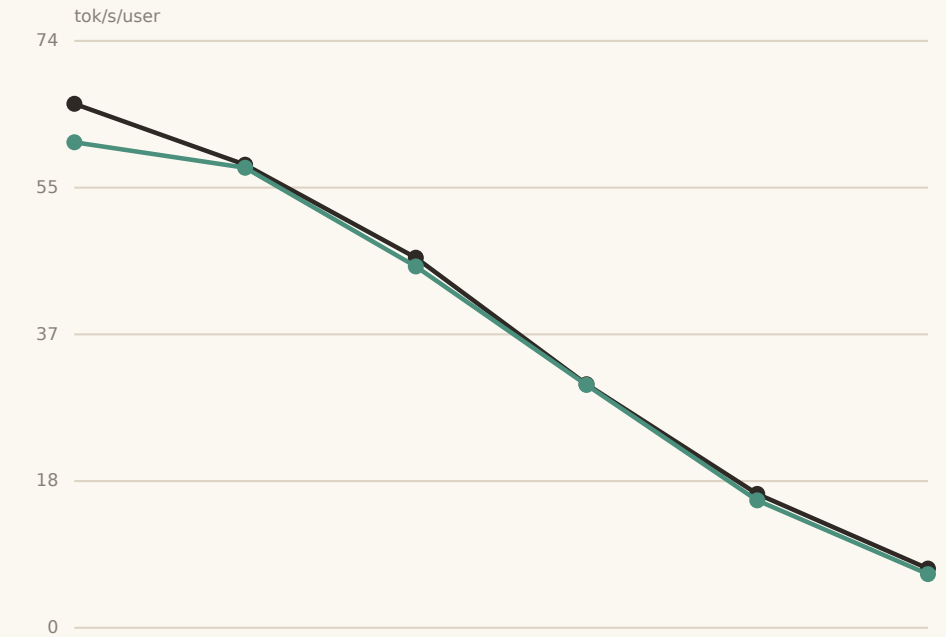
NP1 | Output 300



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GPT-OSS 20B

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KEY TAKEAWAY At 32K, AI MAX+ 395 V5 vs 392: Gemma +14.1%; GPT-OSS -9.3%.

SUMMARY

Observed impact on latency, decode efficiency, and throughput

AI MAX+ 392 → 395 V5 · NP1 · Output 300 · focus on 32K

TTFT

ms

Gemma 4 26B	202,915 → 178,107	+12.2%
GPT-OSS 20B	35,559 → 39,859	-12.1%

TPOT

ms/token

Gemma 4 26B	31.35 → 27.91	+11.0%
GPT-OSS 20B	16.27 → 15.75	+3.2%

Throughput (Aggregate)

tok/s

Gemma 4 26B	1.41 → 1.61	+14.1%
GPT-OSS 20B	7.42 → 6.73	-9.3%

Throughput (Normalized)

tok/s/user

Gemma 4 26B	1.41 → 1.61	+14.1%
GPT-OSS 20B	7.42 → 6.73	-9.3%

Key interpretation

- After aligning Context and Performance mode, AI MAX+ 395 V5 is much closer to AI MAX+ 392 than in the earlier comparison.
- Gemma 32K is now faster on 395 V5: TTFT +12.2%, TPOT +11.0%, throughput +14.1%.
- GPT-OSS remains mixed: decode TPOT is +3.2% at 32K, while TTFT is -12.1% and throughput is -9.3%.
- Driver and Vulkan memory still differ between machines, so remaining deltas should not be presented as pure processor gain.